D.K.T.E. Society's
Textile and Engineering Institute,
Rajwada, Ichalkaranji- 416115

(An Empowered Autonomous Institute Affiliated to Shivaji University, Kolhapur)

NAAC Accredited with A+ Grade, ISO 9001:2015 Certified

DEPARTMENT: Computer Science and Engineering
CURRICULUM
Second Year B.Tech. Engineering Program

With effect from 2024-25

BOS Chairman

Dean Academics

Director

Teaching and Evaluation Scheme for Year 2024-25

Program: B. Tech. Computer Sc. & Engineering

Class.: Second Year, B. Tech. **Semester: SEM-III** **NCrF Level : 5.0** **W.E.F.: 2024-2025**

Sr. No.	Course Code	Course Title	Course Category	Teaching Scheme				Course Credits	Evaluation scheme					
									Theory			Practical		Total
									CIE		SEE	CIE	SEE	
				L	T	P	Contact Hrs/wk		SE-I	SE-II				
01	01CSL201	Mathematics for Computer Science	PCC	3	-	-	3	3	25	25	50	-	-	100
02	01CSL202	Operating Systems	PCC	3	-	-	3	3	25	25	50	-	-	100
03	01CSL203	Data Structure	PCC	3	-	-	3	3	25	25	50	-	-	100
04	01CSL204	Microprocessors	PCC	2	-	-	2	2	25	25	50	-	-	100
05	MDM-I	MDM-I	MDM	2	-	-	2	2	25	25	50	-	-	100
06	01CSL251	Design, Technology and Innovation	HSSM	2	-		2	2	25	25	50			100
07	01CSP205	Data Structure Lab	PCC	-	-	4	4	2	-	-	-	50	50	100
08	01CSP206	Object-Oriented Programming in C++ Lab	PCC	2	-	2	4	3	-	-	-	50	50	100
09	01CSP207	Microprocessors Lab	PCC	-	-	2	2	1	-	-	-	50	50	100
10	MDM-I Lab	MDM-I Lab	MDM			2	2	1	-	-	-	50	-	50
11	01DEL251	Environmental Studies	VEC	1	1	-	2	2	25	25	50	-	-	100
Total				**18**	**1**	**10**	**29**	**24**	**175**	**175**	**350**	**200**	**150**	**1050**

L- Lecture T-Tutorial P-Practical SE-I: Semester Examination-I SE-II: Semester Examination-II CIE – Continuous in Semester Evaluation SEE- Semester End Examination

	BSC/ESC		Program Courses		Multidisciplinary Courses		Skill Courses	Humanities Social Science and Management (HSSM)				Experiential Learning Courses				Liberal Learning Courses
Course Category	**BSC**	**ESC**	**PCC**	**PEC**	**MDM**	**OE**	**VSEC**	**AEC**	**Entrp / Mgmt**	**IKS**	**VEC**	**RM**	**CEP/FP**	**Proj**	**Int/ OJT**	**CC**
Credits			**17**		**03**				**02**		**02**					
Cum. Sum	**14**	**13**	**20**		**03**		**04**	**03**	**02**	**02**	**02**					**02**

Teaching and Evaluation Scheme for Year 2024-25

Program: B. Tech. Computer Sc. & Engineering

Class.: Second Year, B. Tech. Semester: SEM-IV NCrF Level : 5.0 W.E.F.: 2024-2025

Sr. No.	Course Code	Course Title	Course Category	Teaching scheme				Course Credits	Evaluation scheme					
									Theory			Practical		Total
									CIE		SEE	CIE	SEE	
				L	T	P	Contact Hrs/wk		SE-I	SE-II				
01	01CSL208	Software Engineering	PCC	3	-	-	3	3	25	25	50	-	-	100
02	01CSL209	Computer Algorithm	PCC	3	-		3	3	25	25	50	-	-	100
03	01CSL210	Theory of Computer Science	PCC	3	-	-	3	3	25	25	50	-	-	100
04	MDM-II	MDM-II	MDM	2	-		2	2	25	25	50			100
05	01CSL252	Information Technology and Business Management	HSSM	2	-	-	2	2	25	25	50	-	-	100
06	OE-I	Open Elective	OE	3	-	-	3	3	25	25	50			100
07	01CSP253	Modern Indian Language	AEC	1	-	-	1	1	-	-	-	50	-	50
08	01CSP211	Computer Algorithm Lab	PCC	-	-	2	2	1				50	50	100
09	01CSP212	Application Development using Java	VSEC	2	-	2	4	3	-	-	-	50	50	100
10	MDM-II Lab	MDM-II Lab	MDM	-	-	2	2	1	-	-	-	50	-	50
11	01CSP254	Green Computing	VEC	2	-	-	2	2	-	-	-	50	-	50
12	01CSD213	Mini Project	ELC	-	-	4	4	2	-	-	-	50	50	100
13	01CST214	Aptitude Training	AEC	-	-	2	2	1	-	-	-	50	-	50
Total				**21**	**-**	**12**	**33**	**27**	**150**	**150**	**300**	**350**	**150**	**1100**

L- Lecture T-Tutorial P-Practical SE-I: Semester Examination-I SE-II: Semester Examination-II CIE – Continuous in Semester Evaluation SEE- Semester End Examination

	BSC/ESC		Program Courses		Multidisciplinary Courses		Skill Courses	Humanities Social Science and Management (HSSM)				Experiential Learning Courses				Liberal Learning Courses
Course Category	**BSC**	**ESC**	**PCC**	**PEC**	**MDM**	**OE**	**VSEC**	**AEC**	**Entrp / Mgmt**	**IKS**	**VEC**	**RM**	**CEP/FP**	**Proj**	**Int/ OJT**	**CC**
Credits			**10**		**03**	**03**	**03**	**02**	**02**		**02**			**02**		
Cum. Sum	**14**	**13**	**30**		**06**	**03**	**07**	**05**	**04**	**02**	**04**			**02**		**02**

Open Electives:

Sr. No.	Course Code	Course Name
1	01OE201	Financial Management
2	01OE202	Stress Management
3	01OE203	Financial & Managerial Accounting
4	01OE204	Financial Planning and Control
5	01OE205	Supply Chain Management
6	01OE206	Production Planning and Control
7	01OE207	Quality Management

Multidisciplinary Minor (MDM)

- Students should select any one minor, excluding those offered by their parent department, from the diverse range of minors available.
- Students should complete all courses within their selected minor in order to earn credits.

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
Fabric Manufacturing Technology	1	01TTMDL1201	General Textile Technology	III	Textile Technology
	2	01TTMDP1202	General Textile Technology - Lab	III	
	3	01TTMDL1203	Woven Fabric Technology	IV	
	4	01TTMDP1204	Woven Fabric Technology - Lab	IV	
	5	01TTMDL1301	Knitted Fabric Technology	V	
	6	01TTMDL1302	Knitted Fabric Technology - Lab	V	
	7	01TTMDL1303	Nonwoven Fabric Technology	VI	
	8	01TTMDL1304	Nonwoven Fabric Technology - Lab	VI	
	9	01TTMDD1401	Capstone Project	VII	
High Performance Textiles	1	01TMMDL1201	Fundamentals of High Performance Textiles	III	Man Made Textile Technology
	2	01TMMDP1202	Fundamentals of High Performance Textiles Lab	III	
	3	01TMMDL1203	High Performance Fibres	IV	
	4	01TMMDP1204	High Performance Fibres Lab	IV	
	5	01TMMDL1301	Manufacturing of High Performance Textiles	V	
	6	01TMMDP1302	Manufacturing of High Performance Textiles Lab	V	
	7	01TMMDL1303	Automotive and Geo Textiles	VI	
	8	01TMMDP1304	Automotive and Geo Textiles Lab	VI	
	9	01TMMDD1401	Capstone Project	VII	
Machine Maintenance	1	01MEMDL1201	Applied Electronics	III	Textile Plant Engineering
	2	01MEMDP1202	Applied Electronics Lab	III	
	3	01MEMDL1203	Computer Programing	IV	
	4	01MEMDP1204	Computer Programing Lab	IV	
	5	01MEMDL1301	Fluid Flow Systems and Control	V	
	6	01MEMDL1302	Fluid Flow Systems and Control Lab	V	
	7	01MEMDL1303	Mechatronics	VI	
	8	01MEMDT1304	Mechatronics Lab	VI	
	9	01MEMDD1401	Capstone Project	VII	
Chemical Processing of Textiles	1	01TCMDL1201	Introduction to Textile Substrates	III	Textile Chemistry
	2	01TCMDP1202	Introduction to Textile Substrates Lab	III	
	3	01TCMDL1203	Pretreatment of Textiles	IV	
	4	01TCMDP1204	Pretreatment of Textiles Lab	IV	
	5	01TCMDL1301	Colouration of Textiles	V	
	6	01TCMDP1302	Colouration of Textiles Lab	V	
	7	01TCMDL1303	Finishing & Care of Textiles	VI	
	8	01TCMDP1304	Finishing & Care Textiles Lab	VI	
	9	01TCMDD1401	Capstone Project	VII	
Garment Technology	1	01TFMDL1201	Fundamentals of Clothing Manufacturing	III	Fashion Technology
	2	01TFMDP1202	Fundamentals of Clothing Manufacturing Lab	III	
	3	01TFMDL1203	Garment Manufacturing	IV	
	4	01TFMDP1204	Garment Manufacturing Lab	IV	
	5	01TFMDL1301	Design & Pattern Engineering	V	
	6	01TFMDP1302	Design & Pattern Engineering Lab	V	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
	7	01TFMDL1303	Fashion and Garment construction	VI	
	8	01TFMDP1304	Fashion and Garment construction Lab	VI	
	9	01TFMDD1401	Capstone Project	VII	
Embedded Systems	1	01ETMDL1201	Microcontroller	III	Electronics and Telecommunication
	2	01ETMDP1202	Microcontroller Lab	III	
	3	01ETMDL1203	Microcontroller based System	IV	
	4	01ETMDP1204	Microcontroller based System Lab	IV	
	5	01ETMDL1301	Embedded System Design	V	
	6	01ETMDP1302	Embedded System Design Lab	V	
	7	01ETMDL1303	Real Time Embedded System	VI	
	8	01ETMDP1304	Real Time Embedded System Lab	VI	
	9	01ETMDD1401	Capstone Project	VII	
Industrial IOT	1	01ETMDL2201	Sensor and Actuators	III	Electronics and Telecommunication
	2	01ETMDP2202	Sensor and Actuators Lab	III	
	3	01ETMDL2203	Introduction to IOT	IV	
	4	01ETMDP2204	Introduction to IOT Lab	IV	
	5	01ETMDL2301	IOT implementation using Raspberry Pi Boards	V	
	6	01ETMDP2302	IOT implementation using Raspberry Pi Boards Lab	V	
	7	01ETMDL2303	IOT System design	VI	
	8	01ETMDP2304	IOT System design Lab	VI	
	9	01ETMDD2401	Capstone Project	VII	
Industrial Electronics	1	01ETMDL3201	Digital Electronics	III	Electronics and Telecommunication
	2	01ETMDP3202	Digital Electronics Lab	III	
	3	01ETMDL3203	Control System	IV	
	4	01ETMDP3204	Control System Lab	IV	
	5	01ETMDL3301	Microcontroller	V	
	6	01ETMDP3302	Microcontroller Lab	V	
	7	01ETMDL3303	Power Electronics and Drives	VI	
	8	01ETMDP3304	Power Electronics and Drives Lab	VI	
	9	01ETMDD4401	Capstone Project	VII	
Digital System Engineering	1	01ETMDL4201	Basics of Electronics	III	Electronics and Telecommunication
	2	01ETMDP4202	Basics of Electronics Lab	III	
	3	01ETMDL4203	Computer Programming	IV	
	4	01ETMDP4204	Computer Programming Lab	IV	
	5	01ETMDL4301	Applied Electronics and SAP	V	
	6	01ETMDP4302	Applied Electronics and SAP Lab	V	
	7	01ETMDL4303	AI and Data Science	VI	
	8	01ETMDP4304	AI and Data Science Lab	VI	
	9	01ETMDD4401	Capstone Project	VII	
Industrial Automation	1	01MEMDL2201	Introduction to Automation	III	Mechanical
	2	01MEMDP2202	Introduction to Automation Lab	III	
	3	01MEMDL2203	Mechatronics	IV	
	4	01MEMDP2204	Mechatronics Lab	IV	
	5	01MEMDL2301	Industrial Robotics	V	
	6	01MEMDP2302	Industrial Robotics Lab	V	
	7	01MEMDL2303	Control Systems	VI	
	8	01MEMDP2304	Control Systems Lab	VI	
	9	01MEMDP2401	Capstone Project	VII	
	1	01MEMDL1201	Introduction to Product Design	III	Mechanical
	2	01MEMDP1202	Introduction to Product Design Lab	III	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
Product Design and Development	3	01MEMDL1203	Materials and Manufacturing Processes for Product Design	IV	
	4	01MEMDP1204	Materials and Manufacturing Processes for Product Design Lab	IV	
	5	01MEMDL1301	CAD and Digital Prototyping	V	
	6	01MEMDP1302	CAD and Digital Prototyping Lab	V	
	7	01MEMDL1303	Product Lifecycle Management	VI	
	8	01MEMDP1304	Product Lifecycle Management Lab	VI	
	9	01MEMDP1401	Capstone Project	VII	
Energy Engineering	1	01MEMDL3201	Thermal Science	III	Mechanical
	2	01MEMDP3202	Thermal Science Lab	III	
	3	01MEMDL3203	Heat Transfer	IV	
	4	01MEMDP3204	Heat Transfer Lab	IV	
	5	01MEMDL3301	Chemical Engineering Operations	V	
	6	01MEMDP3302	Chemical Engineering Operations Lab	V	
	7	01MEMDL3303	Fluid Mechanics and Machines	VI	
	8	01MEMDP3304	Fluid Mechanics and Machines Lab	VI	
	9	01MEMDD3401	Capstone Project	VII	
Automation and Robotics	1	01MEMDL4201	Applied Electronics	III	Mechanical
	2	01MEMDP4202	Applied Electronics Lab	III	
	3	01MEMDL4203	Computer Programming	IV	
	4	01MEMDP4204	Computer Programming Lab	IV	
	5	01MEMDL4301	Fluid Flow Systems and Control	V	
	6	01MEMDP4302	Fluid Flow Systems and Control Lab	V	
	7	01MEMDL4303	Mechatronics and Robotics	VI	
	8	01MEMDP4304	Mechatronics and Robotics Lab	VI	
	9	01MEMDD4401	Capstone Project	VII	
Software Engineering	1	01CSMDL1201	Software Engineering Processes	III	Computer Science and Engineering
	2	01CSMDL1202	Object Oriented Design and Programming	IV	
	3	01CSMDP1203	Java Programming	IV	
	4	01CSMDL1301	Database Engineering	V	
	5	01CSMDP1302	Database Engineering Lab	V	
	6	01CSMDL1303	Software Testing	VI	
	7	01CSMDP1304	Software Testing Lab	VI	
	8	01CSMDD1401	Capstone Project	VII	
Web Technology	1	01CSMDL2201	Fundamentals of Computer Networks and Database Systems	III	
	2	01CSMDP2202	Computer Networks and Database Systems Lab	III	
	3	01CSMDP2203	Web Page Designing Lab	IV	
	4	01CSMDP2301	Client-Side Programming Lab	V	
	5	01CSMDP2302	Server Side Programming Lab	VI	
	6	01CSMDD2401	Capstone Project	VII	
Digital Design and Manufacturing Systems	1	01CSMDL3201	Basics of Electronic Systems	III	
	2	01CSMDP3202	Basics of Electronic Systems Lab	III	
	3	01CSMDL3203	Digital Illustration Techniques	IV	
	4	01CSMDP3204	Digital Illustration Techniques Lab	IV	
	5	01CSMDL3301	Programming techniques and applications	V	
	6	01CSMDP3302	Programming techniques and applications Lab	V	
	7	01CSMDL3303	CAD CAM Applications	VI	
	8	01CSMDP3304	CAD CAM Applications Lab	VI	
	9	01CSMDD3401	Capstone Project	VII	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
Artificial Intelligence and Machine Learning	1	01AMMDL1201	Introduction to AI-ML	III	Computer Science and Engineering (Artificial Intelligence and Machine Learning)
	2	01AMMDP1202	Basics of AI-ML Lab	III	
	3	01AMMDL1203	Programming with Python for Machine Learning	IV	
	4	01AMMDL1301	Machine Learning Foundations	V	
	5	01AMMDP1302	Machine Learning Foundations Lab	V	
	6	01AMMDL1303	Ethics and Applications in AI	VI	
	7	01AMMDP1304	Deep Learning Lab	VI	
	8	01AMMDD1401	Capstone Project on AI-ML	VII	
Augmented Reality and Virtual Reality	1	01AMMDL2201	Foundations of AR-VR	III	
	2	1AMMDP2202	Basic AR-VR Lab	III	
	3	01AMMDL2203	Programming for AR-VR and 3D fundamentals	IV	
	4	01AMMDP2204	Programming for AR-VR and 3D fundamentals	IV	
	5	01AMMDL2301	3D Modeling and Animation	V	
	6	01AMMDL2302	Virtual Reality Development	VI	
	7	01AMMDD2401	Capstone Project	VII	
Data Analytics	1	01ADMDL1201	Foundation of Data Analysis	III	Artificial Intelligence and Data Science
	2	01ADMDP1202	Foundation of Data Analysis Lab	III	
	3	01ADMDL1203	Exploratory Data Analysis	IV	
	4	01ADMDP1204	Exploratory Data Analysis Lab	IV	
	5	01ADMDL1301	Data Analysis Techniques	V	
	6	01ADMDP1302	Data Analysis Techniques Lab	V	
	7	01ADMDL1303	Data Visualization Techniques	VI	
	8	01ADMDP1304	Data Visualization Techniques Lab	VI	
	9	01ADMDP1401	Capstone Project	VII	
Construction Technology	1	01CEMDL1201	Building Materials & Construction	III	Civil
	2	01CEMDP1202	Building Materials & Construction Lab	III	
	3	01CEMDL1203	Metrics in Civil Engineering	IV	
	4	01CEMDP1204	Metrics in Civil Engineering Lab	IV	
	5	01CEMDL1305	Estimation & Valuation	V	
	6	01CEMDP1306	Estimation & Valuation Lab	V	
	7	01CEMDL1307	Project Planning & Management	VI	
	8	01CEMDP1308	Project Planning & Management Lab	VI	
	9	01CEMDD1409	Capstone Project	VII	
Industrial Electrical System	1	01EEMDL12	Electrical Installation System	III	Electrical
	2	01EEMDP1202	Electrical Installation System Lab	III	
	3	01EEMDL1203	Electrical Switchgears and Safety	IV	
	4	01EEMDP1204	Electrical Switchgears and Safety Lab	IV	
	5	01EEMDL1305	Renewable Energy and Cogeneration in Industry	V	
	6	01EEMDP1306	Renewable Energy and Cogeneration in Industry Lab	V	
	7	01EEMDL1307	Electrical Vehicle Technology	VI	
	8	01EEMDP1308	Electrical Vehicle Technology Lab	VI	
	9	01EEMDD1409	Capstone Project	VII	

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSL201**		**Course Name**	**Mathematics For Computer Science**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	Basic concepts of statistics & the theory of dispersion.
02	Hypothesis testing for inference about the data
03	Theory of combinatorial analysis & probability to solve the problems.
04	Theory of mathematical logic to make the inference.

Course Outcomes:

At the end of the course, students will be able to

01	Describe the basic concepts of mathematics.
02	Apply the principle of statistics to the real-life problem
03	Analyze real-world problems using mathematical principles.

Course Contents

Unit I	Introduction to statistics	6 Hours

Definitions of Population, Variable, Attribute, Census Survey, Sample Survey, Random sample. Raw statistical data, collection, classification, Frequency distribution, class limits & boundary, class width, mid-point. Histogram, Frequency polygon, Frequency curve. Measures of central tendency: Arithmetic Mean (A.M.), Median, Mode, Combined Mean.

Unit II	Measures of dispersion	5 Hours

Range, Quartile deviation, Mean deviation Standard deviation as Absolute measures of dispersion, Coefficientof range, quartile deviation, mean deviation, coefficient of variation as Relative measures of dispersion, coefficient of variation as Relative measures of dispersion, consistency of data.

Unit III	Testing of Hypothesis and Large Sample Tests	8 Hours

Introduction, Hypothesis, Statistic, Critical Region, Errors in Testing, Level of Significance. Test for population mean, equality of population means population proportion & equality of population proportions.

Unit IV	Probability	5 Hours

Counting Principle, Rule of Sum & Product, Random Experiments, Sample space, Events, Concept of Probability, Conditional Probability, Independent events, Random Variables, Probability Distribution Function, Bayes' Theorem.

Unit V	Combinatorial Analysis	6 Hours

First counting principle, second counting principle, permutation, combination, Pigeonhole principle.

Unit VI	Mathematical Logic	9 Hours

Introduction, Statements and notations, Connectives, Statement formulas and truth tables, Well-formed formulas, Tautologies, Equivalence of formulas, Duality law, Tautological implications, functionally complete sets of connectives, Normal Forms, completely parenthesized infix and polish notations, Theory ofInference for statement calculus – validity using truth table, rules of inference, consistency of premises and indirect method of proof.

Texts Books:

1	Statistics for Business and Economics-David R. Anderson, Dennis J. Sweeney,Thomas A. Williams, Jeffrey D. Camm, James J. Cochran- Thirteenth Edition, Cengage Learning Publication
2	Discrete Mathematics and its Applications - Kenneth H. Rosen (AT&T Bell Labs)
3	Mathematical Statistics - J.Fruend.
4	Applied Statistics & Probability of Engineers - Montgomeri & Runger
Reference Books:	
1	Discrete Mathematics - Semyour Lipschutz, MarcLipson (MGH), Schaum's outlines
2	Probability and Statistics- John Schiller, Murray R. Spigel (MGH), Schaum's outlines 3. Probability
Supplementary Readings:	
1. https://www.iith.ac.in/~aravind/Files-DM/LLM-MFCS-2004.pdf	

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSL202**	**Course Name**	**Operating Systems**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3			3		25	25	50

Pre-requisites:

Course Objectives:

01	Fundamental concepts of operating systems
02	Process management, process synchronization, deadlocks, CPU scheduling algorithms and page replacement to solve the problems
03	Memory management scheme, storage management and I/O systems

Course Outcomes:

At the end of the course, students will be able to

01	Explain the basic concepts of the operating system
02	Apply the knowledge of process management, process synchronization, deadlocks, CPU scheduling algorithms and page replacement to solve the problems.
03	Analyze the memory management schemes and storage management

Course Contents

Unit I	Introduction to OS and services	7 Hours

Operating System fundamentals, computer system organization, computer system architecture, OS structure, OS operations, process management, computing environments, OS services, user and OS interface, system calls, types of system calls, system boot.

Unit II	Process management	7 Hours

Process concept, Process States, Process Control Block, Inter-process communication, process scheduling:- basic concepts, Scheduling Criteria, Scheduling Algorithms, Real time CPU scheduling

Unit III	Process synchronization and deadlocks	6 Hours

Background, The critical section problem, Peterson's solution, Mutex Locks, Semaphores, Classic problems of synchronization. System model, deadlock characterization, handling deadlocks, deadlock prevention, deadlock avoidance, deadlock detection, recovery from deadlock.

Unit IV	Memory management	7 Hours

Background, swapping, contiguous memory allocation, segmentation, paging, structure of page table. Virtual memory background, demand paging, page replacement, allocation of frames, thrashing

Unit V	Storage management	6 Hours

File concept, access methods, and Directory and disk structure, file system mounting, file sharing, protection.

Unit VI	Input /Output systems	6 Hours

Overview, I/O hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance.

Texts Books:

1	Abraham Silberschatz, Peter B Galvin, Gerg Gagne "Operating System Concepts",Wiley, 14th Edition

Reference Books:	
1	William Stallings, Operating System: Internals and Design Principles, Prentice Hall, ISBN-10: 0-13-380591-3, ISBN-13: 978-0-13-380591-8, 8th Edition
2	Operating system with case studies in Unix, Netware and Windows NT – Achyut S. Godbole (TMGH).
3	"Operating systems: concepts and design" - Milan Milenkovic (TMGH).
4	"Operating Systems: Internals and Design Principles" by William Stallings
Supplementary Readings:	
1. https://archive.nptel.ac.in/courses/106/105/106105214/ 2. https://www.tutorialspoint.com/operating_system/index.asp	

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSL203**	**Course Name**	**Data Structures**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3			3		25	25	50

Pre-requisites:

Course Objectives:

01	Introduce the concept of data structure.
02	Review of the nature of data and its storage requirements.
03	Design of storage and operations for different types of data.
04	Assessment of complexity of data structures.

Course Outcomes:

At the end of the course, students will be able to

01	Describe the basic terminology of Data Structures.
02	Design algorithms to carry out different operations on data structures.
03	Apply data structures to solve a given problem.
04	Analyze the performance of different data structures

Course Contents

Unit I	Linear List	10 Hours

Data types, Abstract Data Types – model and implementation, Algorithm efficiency, General List – operations, List ADT, List implementations -Array-Algorithms and analysis.

Unit II	Linked List	08 Hours

Linked List-ADT, Singly Linked List, Doubly Linked List, Circular Linked List.

Unit III	Stacks, Queues, and Searching	06 Hours

Stacks ADT, Stack Implementation using linked list and array, applications of stack. Queues – operations ADT, implementations, applications, Circular queue, Priority queues
Searching Techniques - Linear and binary search, Hashing – concept, hashing methods, hash collision, hash collision resolution methods.

Unit IV	Trees	06 Hours

Basic tree concepts, binary tree – properties, implementation, traversal, expression tree, Huffman code binary search tree - concept, implementation, traversal, Search, Insertion, deletion.

Unit V	Balanced Trees	06 Hours

AVL Tree - Concept, balancing tree, insertion, deletion, implementation. Heap –concept, operations: reheapup, reheapdown, build heap, insert and delete nodes, B and B++ tree

Unit VI	Graph	08 Hours

Definition and storage, traversal – depth first and breadth first algorithm, Shortest path Warshall's and Dijkshtra algorithm, spanning tree algorithms.

Texts Books:

1	Data Structures: A Pseudocode Approach with C, Richard F. Gilberg & Behrouz A. Forouzan.
2	Data Structures using C – ISRD Group, TMH publication
3	Schaum's Outlines Data Structures – Seymour Lipschutz (MGH)

Reference Books:	
1	Data Structures and Algorithm Analysis in C, 2 Edition, by Weiss, Pearson Education India.
2	Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles, by Narasimha Karumanchi, Careermonk Publications
Supplementary Readings:	
1. https://www.w3schools.com/dsa/index.php	

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSL204**	**Course Name**	**Microprocessors**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
2	---	---	2		25	25	50

Pre-requisites: Digital systems

Course Objectives:

01	Understand basics of microprocessor.
02	Understand instructions of the 8085 microprocessor and write programs using 8085 instructions.
03	Develop skills in designing memory and IO device interfacing with 8085 Microprocessor.

Course Outcomes:

At the end of the course, students will be able to

01	Explain terminology, architecture, instructions and functionality of Microprocessors.
02	Develop assembly language programs using 8085 Instruction set.
03	Design Memory and IO device interfacing with 8085 Microprocessor.

Course Contents

Unit I	Microprocessor Architecture and Microcomputer System	3 Hours

Introduction: Microprocessor Architecture and its operation- Microprocessor initiated operations and bus organization, internal data operation, Peripheral or externally initiated operations, Input and output (I/O) devices, logic devices for interfacing- Tri-State devices, buffer and decoder.

Unit II	8085 Microprocessor Architecture	5 Hours

The 8085 MPU, 8085 microprocessor pinout and signals, Microprocessor communication and bus timing, De-multiplexing address and Data bus, Generating control signals, The 8085 Architecture (block diagram), 8085 machine cycles and bus timing, op-code fetch machine cycle, Memory read and write machine cycle, IO read and Write machine cycle.

Unit III	Introduction to 8085 instructions	6 Hours

The 8085 programming model, instruction format, data format and storage, addressing modes, Flag register, The 8085 instruction Classification -data transfer operations, Arithmetic Operation, Logic operations, Branch operations, Stack and subroutines, interrupts.

Unit IV	Programming the 8085- Assembly Language Programming	5 Hours

Writing and executing assembly language Programs, programming techniques: Looping and counting, stack and subroutine

Unit V	Memory Interfacing	5 Hours

Memory map and addresses, memory word size, memory and instruction fetch, Memory interfacing- basic concept in memory interfacing, address decoding and memory addresses, interfacing circuit.

Unit VI	Introduction to 8086 Microprocessor	4 Hours

Introduction to 8086 microprocessor, architecture of 8086 Microprocessor, register organization, flag register, Memory segmentation, Physical memory organization of 8086, comparison of 8085 with 8086 microprocessor.

Texts Books:

1	Microprocessor Architecture-Programming and applications with 8085-Ramesh Gaonkar (Penram International Publishing Pvt. Ltd.).
2	Advanced Microprocessors and Peripherals, K. M. Bhurchandi and A.K.Ray –Tata Mcgraw Hill.
Reference Books:	
1	MICROPROCESSOR 8085 : ARCHITECTURE, PROGRAMMING, AND INTERFACING, WADHWA, AJAY-PHI publications
Supplementary Readings:	
1.	https://www.youtube.com/watch?v=I78iyzXQrP4 (Working of 8085 Microprocessor animation)
2.	https://onlinecourses.nptel.ac.in/noc23_ee47/preview

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSL251**	**Course Name**	**Design, Technology and Innovation**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
2			2		25	25	50

Course Objectives:	
01	Case Studies of Design
02	Innovation and methods for innovation
03	Systemic Approaches for innovation

Course Outcomes:	
At the end of the course, students will be able to	
01	Explain different case studies of design
02	Describe various concepts of innovation and research
03	Analyze the different case studies

Course Contents		
Unit I	**Case Studies**	**5 Hours**
Jaipur Foot - A classic innovation, User Centred Helmet Design		
Unit II	**Challenges of Reaching users**	**5 Hours**
Challenges of Reaching a Million Users		
Unit III	**Solution Development**	**5 Hours**
Technology to Solution, Collaborative Excellence		
Unit IV	**Innovation and methods**	**5 Hours**
Collaborative Innovation Methods, Learnings from Grassroot Innovation		
Unit V	**Systemic Approach**	**5 Hours**
Systemic Approach to Biomed Innovations		
Unit VI	**Research and Innovation**	**5 Hours**
Research to Innovation, Smartcane for the Blind- A Success Story		

Course Link:

1. https://onlinecourses.nptel.ac.in/noc24_de14/preview

Second Year B. Tech. (CSE) (Semester–III)						
Course Code	**01CSP205**		**Course Name**	**Data structure Lab**		
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	4	2		50	50

Pre-requisites:

Course Objectives:

01	Planning of data storage for different needs
02	Implementation of linear and non-linear data structures.

Course Outcomes:

At the end of the course, students will be able to

01	Explain terminology in data structure
02	Design an algorithm for data structures to solve the problem.
03	Build a program in C Language for data structures.
04	Analyze data structure.

Guidelines for Experiments

A minimum of 15 experiments will be performed on the topics listed below.

- static list data structure
- Singly Linked List
- Doubly Linked List
- Stack and Queue implementation
- Stack and Queue applications
- Searching Techniques
- Tree
- Graph

Supplementary Readings:

1. https://www.w3schools.com/dsa/index.php

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSP206**	**Course Name**	**Object-Oriented Programming in C++**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
2		2	3		50	50	

Pre-requisites: C Programming

Course Objectives:

01	Fundamentals of C++
02	Concepts of object oriented programming
03	Exception handling in C++
04	Generic Programming in C++
05	Problem solving approach using object oriented concepts.

Course Outcomes:

At the end of the course, students will be able to

01	Describe the terminology of Object Oriented Programming and C++
03	Design object-oriented solutions using C++
04	Analyze programs written in C++

Course Contents

Unit I	Introduction	08 Hours

C++ Program Structure, variables, operators, control structure – if, if- else, switch, looping while, do-while, for, C++ keywords. Input/output – I/O streams and standard I/O devices, cin and associated functions, cout and formatted output. User Defined function- declaration, definition & calling function, function call stack and activation records, storage classes, scope rules, function -default arguments. Reference and reference arguments to the function. Pointer variables, new and delete operator, dynamic arrays

Unit II	Object Oriented Programming - Class and Object	07 Hours

Object Oriented fundamentals, Class and object- concept and need, Class declaration, Class members- member variables and functions, access specifiers, UML notations for class, implementation of member functions. Object Declaration, Accessing class members, class scope, accessor and mutator functions, order of public and private members of the class. Constructors, invoking a constructor, constructors and default parameters, array of objects and constructor, destructor. this pointer, static members, constant objects and member function, Data abstraction , structure and class, information hiding

Unit III	Inheritance and Composition	06 Hours

Inheritance – concept, implementation, base classes and derived classes, members in base classes and derived classes, overriding base class members, UML notations for inheritance, constructors of derived and base classes, destructor in derived class, Inheritance as public, protected and private Composition (Aggregation) and association – concept, implementation and UML Notation

Unit IV	Polymorphism	06 Hours

Polymorphism – need, concept, implementation using function overloading, Multiple Inheritance, functionoverriding, virtual function, pure virtual function, abstract classes, Friend function and friend classes, accessing base class functions from derived class objects, accessing derived class functions from base class objects. Operator overloading: -fundamentals of operator overloading, overloading binary operators, overloading unary operator

Unit V	Exceptions Handling and File Processing	06 Hours

Exception handling: Introduction, Handling exceptions within program, C++ mechanism of exception handling, throwing an exception, order of catch blocks, creating exception classes, rethrowing exceptions, exception handling techniques, Standard Library Exception Hierarchy.
File Processing : Introduction, Files and streams, creating and opening a file, file opening modes, Reading data from file, updating file, Random access file – creating and opening a random access file, reading and writing to a random access file, object serialization.

Unit VI	**Template and Standard Template Library (STL)**	**06 Hours**

Template: Introduction, function template, class template,
STL – Introduction, STL containers, Common member functions in STL container, container headers, typedefs, iterators, iterator operations, STL Algorithms.

Guidelines for Experiments

A minimum of 10 Experiments will be performed on the topic listed below.

- Function overloading
- Operator overloading
- Friend function
- Simple inheritance
- Function template
- Multiple inheritance
- Composition
- Class template
- File handling
- Standard Template Library

Texts Books:

1	C++ Programming: From Problem Analysis To Program Design, (5th Edition), - D.S. MALIK, Cengage Learning.
2	C++ How To Program (8th Edition) by Paul deitel, Harvey deitel, Pearson Publication

Reference Books:

1	Object Oriented Programming in C++ (4th Edition) - Robert Lafore, SAMS Publication
2	Effective C++ 55 Specific Ways to Improve Your Programs and Designs, (3rd Edition) - Scott Meyers, Addison Wesley Publication.
3	The C++ Programming Language,(3rd edition) - Bjarne Stroustrup, Pearson Education India Publication.

Supplementary Readings:

1. C++ API Documentation : https://devdocs.io/cpp/
2. C++ API Reference : https://en.cppreference.com/w/
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012811461821513728109 60_shared/overview
4. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012811214431379456870 1_shared/overview

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01CSP207**			**Course Name**	**Microprocessors Lab**		
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	2	1		50	50	

Pre-requisites: Microprocessor

Course Objectives:

01	Instructions of a microprocessor.
02	Assembly language program using 8085 microprocessor instruction set.

Course Outcomes:

At the end of the course, students will be able to

01	Describe fundamental concepts of Microprocessors.
02	Write and executer assembly language program using 8085 microprocessor instruction set.
03	Analyze an assembly language program.

Guidelines for Experiments

A minimum of 08 Experiments will be performed on the topic listed below.

- 8-bit addition and subtraction with i) Register addressing mode. Ii) Direct addressing mode. iii) Indirect addressing mode
- Packing and unpacking BCD Numbers
- Exchange of blocks
- Smallest number identification
- Largest number identification
- Stack and subroutine

Texts Books:

1	Microprocessor Architecture-Programming and Applications with 8085-Ramesh Gaonkar (Penram International) 4th Edition.

Supplementary Readings:

1	https://www.youtube.com/watch?v=I78iyzXQrP4 (Working of 8085 Microprocessor animation)
2	https://onlinecourses.nptel.ac.in/noc23_ee47/preview

Second Year B. Tech. (CSE) (Semester–III)							
Course Code	**01DEL251**		**Course Name**	**Environmental Studies**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
1	1		2		25	25	50

Pre-requisites: --

Course Objectives:

01	To recall fundamental physical and biological principles that govern natural processes.
02	To state the importance of ecological balance for sustainable development.
03	To describe the impacts of developmental activities and mitigation measures and to further understand the environmental policies and regulations.
04	To identify the complex relationships between scientific approaches to environmental issues and political, social, economic, and ethical perspectives on the environment.

Course Outcomes:

At the end of the course, students will be able to

01	Explain the fundamental concepts of environmental science.
02	Describe the importance of natural resources, ecosystem and biodiversity for maintaining ecological balance.
03	Apply the environmental principles to various scenarios in society and industry.
04	Illustrate the environmental issues and sustainable development.

Course Contents

Unit I	Nature of Environmental studies & Natural resources	05 Hours

Multidisciplinary nature of environmental studies with scope, Need for public awareness. Forest resources, Water resources, Mineral resources, Food resources, Energy resources, Land resources, Role of an individual in conservation of natural resources.

Unit II	Ecosystems	05 Hours

Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Food chains, food webs and ecological pyramids. Introduction, types, characteristics features, structure and function of the following Ecosystem: - a) Forest ecosystem, b) Grassland ecosystem, c) Desert ecosystem, d) Aquatic ecosystems

Unit III	Biodiversity and its Conservation	05 Hours

Introduction, Biogeographical classification of India, Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values. India as a mega-diversity nation; Western Ghats as a bio-diversity region; Hot-spots of biodiversity; Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; Endangered and endemic species of India; Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

Unit IV	Environmental Pollution	05 Hours

Definition: Causes, effects and control measures of: a) Air pollution, b) Water pollution, c) Noise pollution. Solid waste Management: Causes, effects and control measures of urban and industrial wastes. • Role of an individual in prevention of pollution. Disaster management.

Unit V	Social Issues and the Environment	05 Hours

From Unsustainable to Sustainable development; Water conservation, rain water harvesting, Environmental ethics: Issue and possible solutions; Global warming, ozone layer depletion, Wasteland reclamation; Consumerism and waste products.		
Unit VI	**Environmental Protection**	**05 Hours**
Environment Protection Act.; Air (Prevention and Control of Pollution) Act.; Water (Prevention and control of Pollution) Act; Wildlife Protection Act; Forest Conservation Act		
SE-I Assessment		
Complete the minimum 5 assignments given on the following topics: • Nature of Environmental studies & Natural resources • Ecosystems • Biodiversity and its Conservation • Environmental Pollution • Social Issues and the Environment		
SE-II Assessment		
Visit to a local area to document environmental assets river / forest / grassland / hill / mountain or Visit to a local polluted site–urban/rural/Industrial/Agricultural or Study of common plants, insects, birds or study of simple ecosystems-ponds, river, hill slope. Write a practical fieldwork project report based on environmental aspects.		
Texts Books:		
1	Environmental Studies :Handbook by Shivaji University, Kolhapur.	
2	Clark R. S., Marine Pollution, Clanderson Press Oxford (TB) Pg No. 6.	
Reference Books:		
1	Cunningham, W. P. Cooper, T. H. Gorhani, E. & Hepworth, M. T. 2001, Environmental Encyclopedia, Jaico Publ. House, Mumbai, 1196p.	
2	De A. K., Environmental Chemistry, Wiley Eastern Ltd.	
3	Down to Earth, Centre for Science and Environment ®	
4	Gleick, H., 1993, Water in crisis, Pacific Institute for Studies in Dev., Environment & security. Stockholm Env. Institute. Oxford Univ. Press 473p.	
5	Hawkins R. e., Encyclopedia of Indian Natural History, Bombay Natural History Society, Bombay	
Supplementary Readings:		
1. https://onlinecourses.nptel.ac.in/noc23_hs155/preview		

Second Year B. Tech. (CSE)(Semester–IV)							
Course Code	**01CSL208**	**Course Name**	**Software Engineering**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3			3		25	25	50

Pre-requisites:

Course Objectives:

01	Process of software engineering life-cycle.
02	Standard industrial practices for software development

Course Outcomes:

At the end of the course, students will be able to

01	Explain different software engineering processes.
02	Specify the requirement of a problem.
03	Apply the software engineering principles to the given problem

Course Contents

Unit I	Introduction to Software Engineering	08 Hours

The nature of Software, Defining the discipline, The Software Process, Software Engineering Practices, A Generic Process Model, Defining the Framework Activities, Identifying Task Set, Process Patterns, Process Assessment And Improvements, Prescriptive Process Model, Specialized Process Models, Unified Process, Personal And Team Process Model

Unit II	Requirements Engineering	07 Hours

Requirements Engineering, Establishing The Groupwork, Eliciting Requirements, Developing Use Cases, Building The Analysis Model, Negotiating Requirements, Requirements Monitoring, Validating Requirements. Requirement Analysis, Scenario Based Modelling, UML Models That Supplements Use Cases, Class Based Methods, Identifying Analysis Classes, Specifying Attributes, Defining Operations, Class Responsibility Collaboration Modeling, Association and Dependencies, Analysis packages, State Representations, Pattern for Requirement Modeling

Unit III	Design Concept	06 Hours

Design Within the Context of Software Engineering, The Design Process, Design Concepts, The Design Model.

Unit IV	Architectural Design	06 Hours

Software Architecture, Architectural Genres, Architectural Styles, Architectural Consideration, Architectural Design, Assessing Alternative Architectural Design, Pattern Based Architectural Review, Architectural Conformance Checking, Agility And Architecture.

Unit V	Software Component and Interface Design	06 Hours

Components, Designing Class-Based Components, Conducting Component-Level Design, Component-Based Development, The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps

Unit VI	Software Testing	7 Hours

A Strategic Approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Test Strategies for Object-Oriented Software, Validation Testing, System Testing, The Art of Debugging, Internal and External Views of Testing, White-Box Testing, Basis Path Testing, Control Structure Testing, Black-Box Testing	
Texts Books:	
1	Software engineering- A practitioner's Approach- Roger Pressman McGraw-Hill International Edition, 8th edition, 2001.
Reference Books:	
1	Software Engineering- Ian Sommerville, Pearson Higher Education, 10th Edition, 2016
2	An Integrated Approach to Software Engineering-Pankaj Jalote, Springer New York, 2nd Edition, 2013.
3	The Unified Modeling Language User Guide- G. Booch, J. Rambaugh, and I. Jacobson, Addison Wesley, 2nd Edition, 2005.
Supplementary Readings:	
1. https://cse.iitkgp.ac.in/~dsamanta/courses/se/index.html 2. https://nptel.ac.in/courses/106/105/106105087/\ 3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview	

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01CSL209**	**Course Name**	**Computer Algorithms**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3			3		25	25	50

Pre-requisites:

Course Objectives:

01	Understand the fundamental principles of algorithms
02	Apply various algorithmic paradigms such as Divide and Conquer, Greedy Method, Dynamic Programming, Backtracking, and Randomized Algorithms to solve computational problems efficiently.
03	Analyze the complexity of algorithms enabling the evaluation of algorithmic performance and scalability.
04	Explore NP-hard and NP-complete problems and develop an understanding of approximation algorithms.

Course Outcomes:

At the end of the course, students will be able to

01	Describe the fundamentals of algorithms.
02	Analyze the efficiency and effectiveness of algorithms using asymptotic notations
03	Apply algorithmic techniques, including recursive algorithms, divide and conquer, and greedy methods, to solve a variety of computational problems.
04	Design and implement dynamic programming and backtracking solutions for different computational problems.

Course Contents

Unit I	**FUNDAMENTALS OF ALGORITHMS**	**05 Hours**

Introduction, Characteristics of algorithms, Pseudo code Conventions, Recursive Algorithms, Performance analysis, Asymptotic notations (O, Ω, Θ), Performance measurement, Randomized Algorithms, Recurrence relations

Unit II	**DIVIDE AND CONQUER**	**06 Hours**

General method, Binary Search, Ternary Search, Finding the Maximum and Minimum, Merge Sort, Quick Sort, Selection sort

Unit III	**THE GREEDY METHOD**	**07 Hours**

General method, Knapsack Problem, Job Sequencing with Deadlines, Optimal Storage on Tapes, Optimal Merge Patterns, Huffman codes, Single Source Shortest Paths.

Unit IV	**DYNAMIC PROGRAMMING**	**07 Hours**

General method, Multistage Graphs, All Pairs Shortest Paths, Optimal Binary Search Trees, 0/1 Knapsack, Reliability Design, Traveling Salesperson Problem, Flow shop scheduling.

Unit V	**BACKTRACKING**	**06 Hours**

General method, N-Queens Problem, Permutation Tree, Sum of Subsets, Graph Coloring, Hamiltonian Cycle, Knapsack Problem.

Unit VI	NP-HARD AND NP-COMPLETE PROBLEMS	06 Hours
Basic Concepts, P, NP, NP Complete, NP Hard, Cook's Theorem, NP Hard Graph problems, NP Hard Scheduling problems, NP Hard Code Generation Problems, Approximation Algorithms		
Texts Books:		
1	Fundamentals of Computer Algorithms- Ellis Horowitz, SartajSahni, Sanguthevar Rajasekaran, (2nd Edition), Universities Press.	
2	Introduction to Algorithms-Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein (3rd Edition), The MIT Press.	
Reference Books:		
1	The Design and Analysis of Computer Algorithms- A. Aho, J. Hopcroft and J. Ullman, (1st Edition) Addison-Wesley.	
2	Introduction to The Design and Analysis of Algorithms-Anany Levitin, (3rd Edition), Pearson.	
Supplementary Readings:		
1. http://personal.kent.edu/~rmuhamma/Algorithms/algorithm.html 2. https://www.ics.uci.edu/~goodrich/teach/cs260P/notes/ 3. https://archive.nptel.ac.in/courses/106/106/106106131/		

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01CSL210**	**Course Name**	**Theory of Computer Science**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3			3		25	25	50

Pre-requisites:

Course Objectives:

01	To introduce students to the mathematical foundations of computation, the theory of formal languages and grammar.
02	To strengthen the student's ability to understand and conduct mathematical proofs for computations
03	To make the students understand the use of automata theory in Compliers & System Programming.
04	To analyze and design finite automata, pushdown automata, grammars & Turing machines

Course Outcomes:

At the end of the course, students will be able to

01	Prove statements using Mathematical Induction.
02	Describe terminology related to Grammar, Languages, Finite Automata, Pushdown Automata and Turing Machine
03	Design Regular Expression and Finite Automata corresponding to Regular language
04	Design Context Free Grammar corresponding to Context Free Language and Regular Language.
05	Design Push Down Automata to solve a given problem
06	Design Turing machines to solve a given problem

Course Contents

Unit I	Proofs and Regular Languages	07 Hours

Types of Proofs, Mathematical Induction and Recursive definitions, Regular expressions & Regular languages, Operations on Regular languages.

Unit II	Finite Automata	07 Hours

Finite automata definition and representation, union, intersection and complement of Regular Languages and their corresponding FA. Mealy and Moor machines.

Unit III	Non-determinism and Kleen's Theorem	06 Hours

Nondeterministic Finite Automata, Nondeterministic Finite Automata with Λ –transitions, Conversion of NFA-Λ to NFA and DFA. Kleen's Theorem. Minimization of FA

Unit IV	Context Free Languages and Grammar	06 Hours

Examples and definition, Regular Grammar, Derivation and ambiguity, An Unambiguous CFG, Union, concatenation kleen * of CFL, Simplified forms and Normal Forms, Pumping Lemma for context free languages, Intersection and complements of context Free Languages

Unit V	Push Down Automata	08 Hours

Definition of Pushdown Automata, Deterministic PDA, Equivalence of CFG's & PDA's, Top down parsing, bottom up parsing

Unit VI	Turing Machines	08 Hours

Definition of Turing Machine as Language acceptors, combining Turing Machines, Computing a function with a TM Variations in Turing Machines: Turing machines with doubly-infinite tapes, more than one tape, Non-deterministic TM and Universal TM.	
Texts Books:	
1	Introduction to Languages & theory of computations—John C. Martin (MGH).
Reference Books:	
1	An Introduction to Formal Languages and Automata- Fifth edition, Peter Linz.
2	Theory of Computer Science Automata, Languages, and Computation- Third edition
3	"Introduction to Automata Theory, Languages and Computation" - John E. Hopcraft, Rajeev Motwani, Jeffrey D. Ullman (Pearson Edition)
Supplementary Readings:	
1. https://www.javatpoint.com/theory-of-automata 2. https://www.tutorialspoint.com/automata_theory/index.htm 3. https://onlinecourses.nptel.ac.in/noc23_cs70/course	

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01CSL252**	**Course Name**	**Information Technology and Business Management**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
2			2		25	25	50

Pre-requisites:

Course Objectives:

01	Understand different aspects of ERP Systems
02	Understand the ERP implementation process
03	Understand various business models of E-commerce
04	Understand e-commerce marketing, online retail strategies and social networks

Course Outcomes:

At the end of the course, students will be able to

01	Explain the basic concepts of the ERP System and its implementation.
02	Describe various business models of E-commerce.
03	Illustrate e-commerce marketing, online retail strategies and social networks.

Course Contents

Unit I	Introduction ERP	05 Hours

Overview, Accommodating Variety, Integrated Management Information, Supply Chain and Resource Management, Integrated Data Model, Scope, Technology and Benefits of ERP, Building an MIS, Business as a System, Core Process in a Manufacturing Company, Entities forming data Model in a Manufacturing Company

Unit II	ERP implementation	04 Hours

Overview, Role of Consultants, Vendors and Users, Customization, Precautions, Post Implementation Option, ERP Implementation Methodology, Guidelines for ERP Implementation

Unit III	Introduction to E-Commerce	05 Hours

E-commerce: The Revolution is Just Beginning, A Brief History, E-commerce Business Models: Major Business to Consumer (B2C) Business Models, Major Business to Business (B2B) Business Models, Mobile E-commerce (M-Commerce), How E-commerce changes Business - Strategy, Structure and Process.

Unit IV	Consumer Online	05 Hours

Consumer Online: The Internet Audience and Consumer Behavior, Basic Marketing Concepts, Internet Marketing Technologies, B2C and B2B E-commerce Marketing and Business Strategies, The online Retail Sector, Analyzing the Viability of Online Firms

Unit V	E-commerce in Action	05 Hours

E-commerce in Action: E-Retailing Business Models, Common Themes in Online Retailing.
The Service Sector: Offline and Online, Online Financial Services, Online Travel Services, Online Career Services

Unit VI	Social Networks, Auctions and Portals	04 Hours

Social Networks and Online Communities, Social Network features, Online Auctions-Benefits and types of Auctions, E-commerce Portals

Texts Books:	
1	Enterprise resource planning - Vinod Kumar Garg & N. K. Venkita Krishna ,2nd edition [PHI] (for 1st section)
Reference Books:	
1	E-Commerce: Business, Technology, Society - Kenneth C. Laudon, Thirteenth Edition, Pearson Publication
2	E-Commerce: An Indian perspective - S. J. Joseph, Fifth Edition, PHI Publication
Supplementary Readings:	
1. https://onlinecourses.nptel.ac.in/noc22_mg100/preview	

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01OE203**	**Course Name**	**Financial and Managerial Accounting**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	-	-	3		25	25	50

Course Objectives:

01	To understand the fundamental principles and concepts of financial accounting.
02	To learn the techniques of preparing financial statements.
03	To analyze and interpret financial statements for decision-making.
04	To comprehend the basics of managerial accounting.
05	To understand cost concepts and cost behavior.
06	To analyze and interpret cost data for managerial decision-making.
07	To apply accounting techniques in real-world scenarios.

Course Outcomes:

At the end of the course, students will be able to

01	Apply accounting principles and concepts to prepare financial statements.
02	Analyze financial statements to assess the financial health of an organization.
03	Utilize managerial accounting techniques for cost analysis and decision-making.
04	Apply cost concepts to aid in planning and control activities.
05	Utilize managerial accounting techniques for cost analysis and decision-making.
06	Utilize accounting information for strategic decision-making in computer engineering contexts.

Course Contents

Unit I	**Introduction to Accounting**	**5 Hours**
Definition, purpose, and Overview of accounting, principles, and concepts of accounting, Accounting equation, and double-entry system, Recording transactions: journals and ledgers		
Unit II	**Financial Statements**	**5 Hours**
Trial balance, Income statement preparation, Balance sheet preparation, Cash flow statement basics, Preparation of financial statement implementation in software		
Unit III	**Ratio Analysis**	**5 Hours**
Financial leverage, types, Ratio Analysis, types, use, problems/case studies on ratio analysis		
Unit IV	**Management Accounting and Standard Costing**	**5 Hours**
Management Accounting: Nature, scope and tools of Management Accounting; Management Accounting vs. Financial Accounting, Management Accounting Vs. Cost Accounting, Budgeting and Budgetary Control – Meaning, Objectives, Advantages and Limitations of Budgeting, Types of Budget, Practical Questions on Cash Budget and Flexible Budget, Standard Costing – Meaning and Application, Essentials of standard Costing, Variance Analysis- Material and Labour Cost Variances.		
Unit V	**Cost Accounting and Marginal Costing**	**5 Hours**

Cost Accounting: types of cost; elements of cost, Cost-Volume-Profit Analysis, break-even analysis, Decision involving alternative choices: fixation of selling price, make or buy decision and product mix decision.		
Unit VI	**Project Work**	**5 Hours**
Software development Mini Projects for different accounting implementations- Case Study Trading Account, Profit and loss Account, Balance Sheet, Cost Sheet, CVP analysis and break-even analysis. Tax Computation.		
Texts Books:		
1	"Financial Accounting", by Jerry J. Weygandt, Paul D. Kimmel, and Donald E. Kieso	
2	"Management Accounting", by Ray H. Garrison, Eric W. Noreen, and Peter C. Brewer	
3	Financial and Managerial Accounting, by Weygandt, Kimmel, Kieso	
Reference Books:		
1	Accounting: Financial Accounting, Accounting Principles, and Management Accounting, by Income Mastery.	
2	"Management Accounting", by Khan and Jain, Tata McGraw Hill Publishing	
3	Financial & Managerial Accounting for Beginners, Tim Power,	
4	Introduction to Financial Accounting, by T. Horngren Charles, L. Sundern Gary, A. Elliott John, R. Philbrick Danna.	
5	Advanced Accountancy, R.L.Gupta, M.Radhaswami	
6	An Introduction to Accounting, S.N. Maheswari, S.K.Maheshwari	
7	Cost and Management Accounting, Vikas publications , M N Arora	
Supplementary Readings:		
1	https://onlinecourses.nptel.ac.in/noc21_mg75/preview	
2	https://www.careers360.com/university/indian-institute-of-technology-bombay/financial-accounting-certification-course	

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01CSP211**			**Course Name**	**Computer Algorithms Lab**		
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	2	1		50	50	

Pre-requisites:

Course Objectives:

01	To implement efficient searching and sorting algorithm.
02	To develop solutions for complex problems using dynamic programming and a greedy approach.

Course Outcomes:

At the end of the course, students will be able to

01	Describe the fundamentals of algorithms.
02	Discover solutions to problems using algorithm design paradigms like Divide and Conquer and Greedy Approach.
03	Apply Dynamic Programming and Backtracking Approach to tackle the problems.

Guidelines of Experiments

A minimum of 10 Experiments will be performed on the topic listed below.

- Complexity calculation and analysis.
- Bubble Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Prims Algorithm
- Kruskal Algorithm
- Single Source Shortest Path
- Huffman Code
- Knapsack Problem Instance
- Job sequencing with deadline
- Multistage Graph
- All pair shortest path algorithm
- Optimal Binary Search tree
- 0 / 1 Knapsack Problem Instance
- N-Queens Problem
- Complete one project using above topics.

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01CSP212**			**Course Name**	**App Development using Java Lab**		
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
2		2	3		50	50	

Pre-requisites:

Course Objectives:

01	Understand Java UI Components
02	Design User Interfaces using Object Oriented features
03	Implementing Basic UI Components

Course Outcomes:

At the end of the course, students will be able to

01	Explain fundamental concepts of Java and UI Principles.
02	Design an object-oriented solution for a given problem using Java OOP.
03	Develop a User Interface Application using Java.

Course Contents

Unit I	Introduction	05 Hours

Introduction: Features of Java, The Java Programming Environment JVM, JIT Compiler, Byte Code Concept, A Simple Java Program, Objects and Classes: Object-Oriented Programming Concepts, Declaring Classes, Key principles: encapsulation, inheritance, polymorphism, and abstraction, I/O: Streams, Text input and output, character streams, Reading and writing binary data in to a file.

Unit II	Java UI Classes	05 Hours

Overview of Java UI toolkits (AWT, Swing, JavaFX), JFrame, JPanel, JDialog, JApplet, JButton, JLabel, JTextField, JTextArea, JCheckBox, JComboBox, JList and JComboBox, JTable and JTree, JTabbedPane and JSplitPane, JScrollPane and JViewport, JToolBar and JPopupMenu.

Unit III	Building Java Applications	06 Hours

Understanding Relationships Between Classes, defining associations, aggregations, and compositions, Differences and use cases for each type of relationship, UML diagrams for visualizing class relationships, Association in Java, Aggregation in Java, Composition in Java, Composition in Java, Handling bidirectional associations and aggregations.

Unit IV	Layout Management	05 Hours

Introduction to Layout Management, APIs for Border Layout, panels, Text Input, Choice Components, Menus, Dialog Boxes, Overview of Layout Managers, Introduction to layout managers, Importance of layout management in GUI applications, Types of Layout Managers, FlowLayout, BorderLayout, GridLayout ,CardLayout, GridBagLayout, GroupLayout, BoxLayout, SpringLayout, Custom Layout Managers, Creating custom layout managers, Extending Layout Manager and Layout Manager interfaces, Choosing the Right Layout Manager, Criteria for selecting a layout manager, Comparing different layout managers

Unit V	Event Listener	04 Hours

Event Handling: Event Handling Mechanism in Java, Types of Event Listeners, Implementing Event Listeners, Registering Event Listeners, Handling Common Events, Custom Event Handling, Event Adapter Classes, Event Handling in Swing vs. AWT, Debugging Event Handling Code.

Unit VI	Advanced Applications in Java	05 Hours

Concurrency and Multithreading, Networking and Distributed Applications, Graphics, Database Integration, Advanced Java Frameworks and Libraries, Machine Learning and AI (WEKA)
Guidelines for Experiments
A minimum of 10 Experiments will be performed on the topic listed below. • OOP Concepts like classes, object • Inheritance in UI • Polymorphism in UI • Association and Aggregation • Layout management • Event Listener • Multithreading

Texts Books:	
1	Java How to Program, Early Objects (Deitel: How to Program) 11th Edition by Paul Deitel (Author), Harvey Deitel (Author)
2	"JavaFX 9 by Example" by Carl Dea, Gerrit Grunwald, José Pereda, Sean Phillips
Reference Books:	
1	"Head First Java" by Kathy Sierra and Bert Bates
2	"Java: A Beginner's Guide" by Herbert Schildt
3	"Mastering JavaFX 8 Controls" by Hendrik Ebbers:
Supplementary Readings:	
1. https://docs.oracle.com/javase/tutorial/uiswing/index.html	

Second Year B. Tech. (CSE) (Semester–IV)						
Course Code	**01CSP254**	**Course Name**	**Green Computing**			
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE	SEE Marks
2	-	-	2		50	--

Course Objectives:	
01	To understand the need for green computing.
02	To introduce the green computing framework.
03	To learn case studies of green computing.

Course Outcomes:	
At the end of the course, students will be able to	
01	Describe the fundamentals of Green Computing.
02	Explain the Green Computing Grid Framework.
03	To develop various case studies.

Course Contents		
Unit I	**Green IT: An overview**	**5 Hours**
Green IT fundamentals - Environmental Impacts of IT - Green IT standards - Applying IT for enhancing environmental sustainability.		
Unit II	**Green Devices and Hardware**	**5 Hours**
Life cycle of a device or hardware - Reuse, Recycle and dispose. Green software - Energy saving software techniques, Green information systems, evaluating software impact to platform power.		
Unit III	**Grid Framework**	**5 Hours**
Green grid framework - Materials recycling - Green data center - Virtualization of IT systems - Inter-organizational enterprise activities and Green issues.		
Unit IV	**Managing Green IT**	**5 Hours**
Implementation of Green IT, Information Assurance and communication – Green Enterprise transformation roadmap - Green compliance.		
Unit V	**Law, Standards and Protocols**	**5 Hours**
Regulatory environment and IT manufacturers, Non-regulatory government initiatives, Green building standards, Green data centers.		
Unit VI	**Case Studies**	**5 Hours**
The Environmentally Responsible Business Strategies (ERBS) – Case Study Scenarios for Trial Runs – Case Studies – Applying Green IT Strategies and Applications to a Home, Hospital, Packaging Industry, and Telecom Sector.		

Texts Books:	
1	Bhuvan Unhelkar, —Green IT Strategies and Applications-Using Environmental Intelligence, CRC Press, June 2014.
2	Woody Leonhard, Katherine Murray, —Green Home computing for dummies, August 2012.
Reference Books:	

1	Alin Gales, Michael Schaefer, Mike Ebbers, —Green Data Center: steps for the Journey, Shroff/IBM rebook, 2011.
2	John Lamb, —The Greening of IT, Pearson Education, 2009.
3	Jason Harris, —Green Computing and Green IT- Best Practices on regulations & industry, Lulu.com, 2008
Supplementary Readings:	
1. https://www.youtube.com/watch?v=zz_8F3cenBk	

Second Year B. Tech. (CSE) (Semester–IV)							
Course Code	**01CSD213**	**Course Name**	**Mini Project**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	4	2		50	50	

Pre-requisites:

Course Objectives:

01	To expose the students to use engineering approaches to solve real-time problems
02	To learn the skills of team building and teamwork.
03	To develop logical skills and use appropriate data structures for solving engineering problems and puzzles.

Course Outcomes:

At the end of the course, students will be able to

01	Draft and analyze the requirements of the problem.
02	Design solution for the problem
03	Develop program for the given problem and test the developed program
04	Write report for the project.

Guidelines

The project should be undertaken preferably by a group of 3-4 students who will jointly work and implement the project. The topic for the project must be based upon a societal problem or real-world problem. To select real-life problems students should communicate with their parents and identify the issues from their professional work. The project work should be completed in all aspects of analysis, design, Implementation, and testing (SDLC). The group will select a problem with the approval of the guide and carry out requirements gathering and analysis, requirements specification, design document, coding, test plans; testing, and installation reports (if any) for the selected problem statement. Further, the group will write a report covering the project details and give a presentation. Students also have to maintain a diary of schedule, cost and other managerial activities. All phases of SDLC along with the diary should be considered for evaluation of project.

Reference Books:

1	Software Engineering - Concepts & Practices by Ugrasen Suman (Cenage Learning)
2	Software Engineering Fundamentals -- Behforooz & Hudson
3	Software Engineering by Roger Pressman, McGraw-Hill Publication, 9th Edition
4	An Integrated Approach to software engineering by Pankaj Jalote, Narosa Publication, 3rd Edition
5	Fundamentals of Software Engineering by Rajib Mall, PHI, 3rd Edition.
6	The Unified Modeling Language User Guide by Grady Booch, James Rumbaugh, Ivar Jacobson
7	Software Engineering by Ian Sommerville, Pearson Publication, 9th Edition

Supplementary Readings:

1. http://vlabs.iitkgp.ac.in